

Aleksandr Drozd

Curriculum Vitae

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Tokyo, Japan
✉ alex@blackbird.pw
🌐 blackbird.pw



An executive technologist working at the convergence of Artificial Intelligence, Software Engineering, and High-Performance Computing — combining deep R&D with the delivery of complex production systems. PhD Tokyo Tech → RIKEN → Startup CTO.

Key Competences

AI Research & Innovation — Deep expertise spanning the full evolution of AI: from symbolic methods (GOFAI) and corpus linguistics to static embeddings, to modern Large Language Models and Agentic Systems. Leading cross-functional research teams and translating state-of-the-art research into scalable, production-grade implementations.

Software Architecture & Product Delivery — Over 20 years of hands-on software development. Architecting complex commercial systems—from high-load online trading platforms and industrial machinery control to modern distributed AI products. Currently serving as CTO at an AI startup.

National-Scale Computing & Sovereign AI Infrastructure — Leading technical evaluation and benchmarking for sovereign-grade computing infrastructure, including the successor to the Fugaku Supercomputer. Technical advisor for multi-million dollar hardware procurements (e.g., the 4,000-GPU ABCI supercomputer), navigating vendor relationships to balance cost, power efficiency, and AI performance.

I combine the theoretical depth of a published scientist (H-Index: 18) with the pragmatic rigor of an architect who has shipped industrial-grade systems—and I remain hands-on with the technology while driving strategy, road-mapping, and partner negotiations.

Technical Proficiency

- **Core Languages & HPC:** Python (Expert, 20+ years), C/C++, CUDA, Distributed Computing (MPI, OpenMP). Emerging interests: Go, Rust, Zig.
- **AI & Data Stack:** PyTorch, Large Language Models (LLM) Fine-tuning & Deployment, Agentic Systems, RAG and Data Pipelines.
- **Architecture & Cloud:** Distributed Systems Design, Hybrid Cloud (AWS/On-Prem), Linux Systems Programming, Containerization (Docker/Singularity), Databases (SQL, NoSQL), CI/CD.
- **Rapid Prototyping (Full Stack):** Modern lightweight stacks for fast MVP delivery (Flask, FastAPI, HTMX, Tailwind, SQLite, Go). Capable of full-stack delivery from backend logic to frontend interfaces.
- **Decentralized Technologies:** P2P architectures, blockchain, smart contracts, trustless data layers for autonomous agent swarms and resilient infrastructure.

Employment History

- 2026.04-onwards **CTO** at AI-KIP.
- 2021.07-onwards **Consultant** working with startups and companies on building and integrating AI solutions (via Amigawa GK).
- 2019.07-2026.03 **Research Scientist** at RIKEN Center for Computational Science (R-CCS), High Performance Artificial Intelligence Systems Research Team.
- 2019.07-2026.03 **Visiting Researcher** at Tokyo Institute of Technology, School of Computing
- 2019.02-2019.06 **Visiting Scientist** at RIKEN Center for Computational Science (R-CCS), High Performance Big Data Research Team.
- 2018.06-2019.06 **Invited Researcher** at AIST-Tokyo Tech Real World Big-Data Computation Open Innovation Laboratory (RWBC-OIL).
- 2018.04-2019.06 **Researcher** at Tokyo Institute of Technology, School of Computing, Department of Mathematical and Computing Science.
- 2014.04-2018.03 **Researcher** at Tokyo Institute of Technology, Global Scientific Information and Computing Center.
- 2005.06-2010.05 **Lecturer / Senior Lecturer** (from 2008) at Moscow State University (Sevastopol Branch, <http://sev.msu.ru/>), Department of Computational Mathematics and Cybernetics.
- 2006.09-2009.06 **Software Architect and Developer** at Outsourcing Ukraine (<http://www.outsourcing-ukraine.com/>).
- 2005.05-2006.09 **Software Developer** at Soft-Pilot 2000.

Education

- 2010.04-2014.03 **Ph.D.**, Tokyo Institute of Technology, Graduate School of Information Science and Technology. (obtained 2014.03.26)
- Thesis title: Memory-Conscious Optimizations for Sorting and Sequence Alignment for Massively Parallel Heterogeneous Architectures.
- 2000.09-2005.05 **Specialist degree** (M.Sc. equivalent), Moscow State University. Department of Computational Mathematics and Cybernetics.
- Thesis title: Semantic Pseudo-Code: Approach to Meaning-Base Search.

Language Proficiency

English: fluent

Japanese: fluent

Russian: fluent

Ukrainian: fluent

Recent Projects

Multi-tenant agentic AI platform (KIP) – as CTO. Architecting and building KIP from the ground up: a production, multi-tenant system that autonomously discovers relevant real-world entities and manages each through a full lifecycle – enrichment, continuous monitoring for changes and signals, scoring and prioritization, and context-aware action – with a human in the loop throughout. The interesting part is the engineering: per-tenant knowledge graphs, a staged processing pipeline over a home-grown fair-share task queue, and a shared data-caching layer that keeps external-API spend bounded at scale, with AI orchestrated for reasoning, scoring, and generation. Built cost-first on a modern web + Python stack (React, FastAPI, Neo4j, Redis), fully containerized. (Applied domain: B2B go-to-market automation.)

AI for optimizing program codes. Principal investigator on a competitively-funded JSPS KAKENHI project (grant JP22H03600) developing a machine-driven approach to automate the labor-intensive optimization of scientific software. By searching a space of formally-defined loop transformations (using a polyhedral representation of iteration spaces), the method guarantees correctness of the generated code – unlike end-to-end generation with language models.

Developing next national flagship supercomputer. At RIKEN, I led technical evaluation and AI benchmarking efforts for the successor to supercomputer Fugaku (targeted 2028). This included coordinating deep learning workload benchmarks, surveying emergent trends in AI model development, and evaluating hardware options including simulation of not-yet-existing designs.

In the early stages of supercomputer Fugaku development, I **led an effort to create a deep learning ecosystem** for it. In-depth study of architectures of deep learning frameworks, libraries of specialized performance primitives and intermediate representations allowed us to identify the most promising directions and create large-scale partnership involving Fujitsu Ltd, ARM Ltd, Linaro Ltd. At the later stage of the project, engineering effort was largely moved to Fujitsu Ltd.

During my stay at AIST I was involved in the procurement of the 4,000-GPU “ABCI” supercomputer purpose-built by AIST for AI workloads.

Natural Language Processing: In scientific work one of my focus areas has been natural language processing and computational linguistics: from symbolic methods to embeddings to LLMs. During the era of static word embeddings, I led the development of a library for their training and quality assessment (<https://vecto.space/>). More recently, I worked on interpretability of language models, improving generalization, etc. Finally, I stay passionate about **using and further developing my software development skills**. I introduce best software development practices to research work done by me and collaborators, contribute to open-source project and supervise engineering efforts in some of the research projects within the team.

Commercial software architecture. Across two decades of hands-on development I have architected and shipped a range of production systems – among them a high-load online trading platform for jewelry and gems, and real-time control software for industrial machinery (ported across real-time operating systems). Alongside these, ongoing open-source and side projects, such as a generative-art library in Python (<https://pycontextfree.blackbird.pw/>).

Selected Fellowships and Grants

- JSPS KAKENHI Grant number JP22H03600 adopted FY 2022: "Automated, Scalable, and Machine Learning-Driven Approach for Generating and Optimizing Scientific Application Codes".
- HPCI Project hp210265 "Training Novel Types of Large-Scale Language Models: Tuning". 1000000 node-hours on supercomputer Fugaku.
- 2021 ABCI Grand Challenge #3: 1000 A-100 GPU/days for scalable weakly supervised video representation learning study.
- HPCI Project hp200281 "Training Novel Types of Large-Scale Language Models: Preparation". 100000 node-hours on supercomputer Fugaku.
- JSPS KAKENHI Grant number JP17K12739 adopted FY 2017: "Corpora on Demand: Scalable Methods of Obtaining Linguistic Data".
- Japanese Government (Monbukagakusho) scholarship for conducting PhD research 2010-2014.

Teaching and Supervision

As a research scientist at the RIKEN Center for Computational Science I have been supervising a number of interns admitted to the RIKEN summer internship program, as well as interns invited individually and supported by our or external research funds. Most recently (September to November 2022) I have been supervising (now Dr.) Giovanni Puccetti from the Scuola Normale Superiore, Pisa. His internship resulted in a publication in a prestigious EMNLP conference.

As a post-doctoral appointee at the Tokyo Institute of Technology, I co-advised the PhD work of several students (the main supervisor was prof. Satoshi Matsuoka), most recently Shweta Salaria, thesis title "Cross Architecture Performance Prediction".

Courses taught as a lecturer / senior lecturer at the Moscow State University, Faculty of Computational Mathematics and Cybernetics through years 2005-2010:

- **Operating Systems:** Architecture of Unix-like operating systems, inter-process communication mechanisms, C programming language.
- **Object-Oriented Software Design:** C++ programming language, object-oriented approach to software development.
- **Computer Graphics:** basic 2D drawing, 3D projections and transformations, shading, ray tracing.
- **Parallel Data Processing:** Theoretical foundations of parallel computing, OpenMP and MPI libraries/run-times, GPU computing.

I was responsible for developing curricula and teaching materials for these courses, as well as conducting the final examinations. I have also taught fundamentals of computer science at the Faculty of Philology of the Moscow State University.

Community Service

Organizing Academic Events

- International Workshop on Large Language Models (LLMs) and HPC 2024-2025. <https://llmhpc.github.io/2024/>
- The Workshop on Insights from Negative Results in NLP series (2022 - 2025) In conjunction with *ACL conferences. <https://insights-workshop.github.io>
- Benchmarking in the Data Center: Expanding to the Cloud in conjunction with PPOPP 2022: Principles and Practice of Parallel Programming 2022. <https://parallel.computer>
- International Workshop COmputing using EmeRging EXotic AI-Inspired Systems (CORTEX'22). Co-hosted with IPDPS 2022 conference. <https://cortex.ws>
- "Deep Learning from HPC Perspectives: Opportunities and Challenges" Mini-Symposium at SIAM PP 2018 conference. http://meetings.siam.org/sess/dsp_programsess.cfm?SESSIONCODE=63584
- "The Third Workshop on Evaluating Vector Space Representations for NLP". Co-located with NAACL 2019 conference on June 6 or 7, 2019. <https://repeval2019.github.io>

Chairing/ Program Committee Participation

- I have served as an area chair for the series of EMNLP conferences (efficient NLP track) and program committee member of a number of conferences and workshops, including NAACL, *SEM, SC, ISC, PARCO among others.

Tutorials

- "Text Representation Learning and Compositional Semantics". Tutorial 5 at the 11th Asian Conference on Machine Learning (ACML 2019) <http://www.acml-conf.org/2019/tutorials/drozd-rogers/>
- "Distributional Compositional Semantics in the Age of Word Embeddings: Tasks, Resources and Methodology". Tutorial 4 at LREC 2018 conference. http://text-machine.cs.uml.edu/lrec2018_t4

Misc

- From 2017 I serve as one of the organizers of "Tokyo Machine Learning Gym" meetup.
- I actively participate in local meetups and talkathon, recently giving talks at events such as CrossRoads and Tokyo AI Talks etc.

Publication List

(1670 citations, h-index=18 as of January 2026)

- Taishi Nakamura, Mayank Mishra, Simone Tedeschi, Yekun Chai, Jason T Stillerman, Felix Friedrich, Prateek Yadav, Tanmay Laud, Vu Minh Chien, Terry Yue Zhuo, Diganta Misra, Ben Bogin, Xuan-Son Vu, Marzena Karpinska, Arnav Varma Dantuluri, Wojciech Kusa, Tommaso Furlanello, Rio Yokota, Niklas Muennighoff, Suhas Pai, Tosin Adewumi, Veronika Laippala, Xiaozhe Yao, Adalberto Junior, Alpay Ariyak, Aleksandr Drozd, Jordan Clive, Kshitij Gupta, Liangyu Chen, Qi Sun, Ken Tsui, Noah Persaud, Nour Fahmy, Tianlong Chen, Mohit Bansal, Nicolo Monti, Tai Dang, Ziyang Luo, Tien-Tung Bui, Roberto Navigli, Virendra Mehta, Matthew Blumberg, Victor May, Huu Nguyen, Sampo Pyysalo. **Aurora-M: Open Source Continual Pre-training for Multilingual Language and Code**. COLING 2025.
- Jens Domke, Emil Vatai, Balazs Gerofi, Yuetsu Kodama, Mohamed Wahib, Artur Podobas, Sparsh Mittal, Miquel Pericas, Lingqi Zhang, Peng Chen, Aleksandr Drozd, Satoshi Matsuoka. **At the locus of performance: Quantifying the effects of copious 3D-stacked cache on HPC workloads**. ACM Transactions on Architecture and Code Optimization 20, 2023.
- Satoshi Matsuoka, Jens Domke, Mohamed Wahib, Aleksandr Drozd, Torsten Hoefler. **Myths and legends in high-performance computing**. The International Journal of High Performance Computing Applications 37, 2023.
- Satoshi Matsuoka, Jens Domke, Mohamed Wahib, Aleksandr Drozd, Andrew A Chien, Raymond Bair, Jeffrey S Vetter, John Shalf. **Preparing for the Future—Rethinking Proxy Applications** Computing in Science & Engineering N24 (2), 2022, pp 85-90
- Truong Thao Nguyen, François Trahay, Jens Domke, Aleksandr Drozd, Emil Vatai, Jianwei Liao, Mohamed Wahib, Balazs Gerofi **Why globally re-shuffle? Revisiting data shuffling in large scale deep learning**. 2022 IEEE International Parallel and Distributed Processing Symposium (IPDPS) pp 1085-1096.
- Giovanni Puccetti, Anna Rogers, Aleksandr Drozd and Felice Dell’Orletta. **Outlier Dimensions that Disrupt Transformers are Driven by Frequency** Findings of the Association for Computational Linguistics: EMNLP 2022, pp 1286–1304.
- Prajwal Bhargava, Aleksandr Drozd, Anna Rogers. **Generalization in NLI: Ways (Not) To Go Beyond Simple Heuristics**. Proceedings of the Second Workshop on Insights from Negative Results in NLP (Insights 2021), pp 125–135.
- Steven Farrell, Murali Emani, Jacob Balma, Lukas Drescher, Aleksandr Drozd et al. **MLPerf™ HPC: A Holistic Benchmark Suite for Scientific Machine Learning on HPC Systems**. 2021 IEEE/ACM Workshop on Machine Learning in High Performance Computing Environments (MLHPC)
- Jens Domke, Emil Vatai, Aleksandr Drozd, et al. **Matrix Engines for High Performance Computing: A Paragon of Performance or Grasping at Straws?** IPDPS 2021: International Parallel and Distributed Processing Symposium. pp 1056-1065.
- Mohamed Wahib, Haoyu Zhang, Truong Thao Nguyen, Aleksandr Drozd, Jens Domke, Lingqi Zhang, Ryousei Takano, Satoshi Matsuoka. **Scaling distributed deep learning workloads beyond the memory capacity with KARMA**. Proceedings of SC 20: the

International Conference for High Performance Computing, Networking, Storage and Analysis. Article No.: 19. pp 1–15.

- Shweta Salaria, [Aleksandr Drozd](#), Artur Podobas, Satoshi Matsuoka. **Learning Neural Representations for Predicting GPU Performance**. International Conference on High Performance Computing 2019, pp 40–58.
- Marzena Karpinska, Bofang Li, Anna Rogers and [Aleksandr Drozd](#). **Subcharacter Information in Japanese Embeddings: When Is It Worth It?** In Proceedings of the Workshop on Relevance of Linguistic Structure in Neural Architectures for NLP (RELNLP) 2018 at ACL 2018. Melbourne, Australia. pp 28–37.
- Bofang Li and [Aleksandr Drozd](#). **Subword-Level Composition Functions for Learning Word Embeddings**. *Proceedings of The 2nd Workshop on Subword and Character level models in NLP (SCLeM)* at NAACL 2018. pp 38–48.
- Shweta Salaria, Aleksandr Drozd, Artur Podobas, Satoshi Matsuoka. **Predicting performance using collaborative filtering** 2018 IEEE International Conference on Cluster Computing (CLUSTER), pp 504–514
- Anna Rogers, [Aleksandr Drozd](#) and Bofang Li. **The (too Many) Problems of Analogical Reasoning with Word Vectors**. In Proceedings of the 6th Joint Conference on Lexical and Computational Semantics (*SEM 2017), Association for Computational Linguistics, pp 135–148, Vancouver, Canada.
- [Aleksandr Drozd](#), Anna Gladkova, Satoshi Matsuoka. **Word Embeddings, Analogies, and Machine Learning: Beyond King - Man + Woman = Queen**. Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers, pp 3519–3530, Osaka, Japan, December 11-17 2016
- Mateusz Bysiek, [Aleksandr Drozd](#) and Satoshi Matsuoka. **Migrating Legacy Fortran to Python While Retaining Fortran-Level Performance through Transpilation and Type Hints**. Proceedings of PyHPC 16: the 6th Workshop on Python for High-Performance and Scientific Computing. pp 9-18.
- Anna Gladkova and [Aleksandr Drozd](#). **Intrinsic Evaluations of Word Embeddings: What Can We Do Better?** in Proceedings of The 1st Workshop on Evaluating Vector Space Representations for NLP, Berlin, Germany, 2016, pp. 36–42.
- Anna Gladkova, [Aleksandr Drozd](#) and Satoshi Matsuoka. **Analogy-based Detection of Morphological and Semantic Relations With Word Embeddings: What Works and What Doesn't**. Proceedings of NAACL-HLT-SRW 2016, pp 8–15.
- [Aleksandr Drozd](#), Anna Gladkova, Satoshi Matsuoka. **Discovering Aspectual Classes of Russian Verbs in Untagged Large Corpora**. The 2015 IEEE International Conference on Data Science and Data Intensive Systems (DSDIS 2015), At Sydney, Australia, Dec 2015, pp 61 - 68.
- [Aleksandr Drozd](#), Anna Gladkova, Satoshi Matsuoka. **Python, Performance and Natural Language Processing**. 5th Workshop on Python for High-Performance and Scientific Computing, at Austin, Texas, USA, Nov 2015 in conjunction with SC15, pp 1-10.
- [Aleksandr Drozd](#), Olaf Witkowski, Satoshi Matsuoka, Takashi Ikegami. **Signal-Driven Swarming: A Parallel Implementation of Evolved Autonomous Agents to Perform A**

Foraging Task. Proceedings of SWARM 2015 - The First International Symposium on Swarm Behavior and Bio-Inspired Robotics, Kyoto, Oct 2015.

- Aleksandr Drozd, Naoya Maruyama and Satoshi Matsuoka. **Sequence Alignment on Massively Parallel Heterogeneous Systems**, IEEE 26th International Parallel and Distributed Processing Symposium Workshops & PhD Forum. 2012, Shanghai, China. Proceedings of IPDPS 12 workshops, pages 2498 - 2501
- Aleksandr Drozd and Satoshi Matsuoka. **A Multi GPU Read Alignment Algorithm with Model-based Performance Optimization**, 10th International Conference, on High Performance Computing for Computational Science - VECPAR 2012, Kobe, Japan, July 17-20, printed as Springer's Lecture Notes in Computer Science N7851, pages 270-277.
- Bofang Li, Aleksandr Drozd, Yuhe Guo, Tao Liu, Satoshi Matsuoka, Xiaoyong Du. **Scaling Word2Vec on Big Corpus** Data Science and Engineering, June 2019, Volume 4, Issue 2, pp 157–175.
- Hideyuki Shamoto, Koichi Shirahata, Aleksandr Drozd, Hitoshi Sato and Satoshi Matsuoka. **GPU-Accelerated Large-Scale Distributed Sorting Coping with Device Memory Capacity**. IEEE Trans. Big Data 2(1): 57-69 (2016)
- Aleksandr Drozd, Olaf Witkowski, Satoshi Matsuoka and Takashi Ikegami. **Critical Mass in the Emergence of Collective Intelligence: a Parallelized Simulation of Swarms in Noisy Environments**. Artificial Life and Robotics 2016, volume 21, number 3, pp 317-323
- Anna Gladkova and Aleksandr Drozd, **Towards Easier Querying of XML-based Linguistic Corpora**, Taurida Bulletin of Mathematics and Informatics. #2, 2009, pages 71-77 <http://tvim.info/node/146>